



brains to think about their overall problem, get their propositions and their market right in one country before they think of going global. “Globalisation needs to be thought of on the basis of the product that you’re building and the complexities that you get into but, most importantly, on the core IP that you’re able to demonstrate which is not available in other countries,” he adds.

Pricing is another crucial aspect that needs to be taken care of, especially when expanding to more than one region, because one price doesn’t fit all. For instance, the pricing of cost-sensitive markets like Southeast Asia and South America cannot be the same as that of, say, North America. “We are fundamentally talking about a different marketing effort in these countries and set of sales resources in each of these countries, and for me intuitively it means that it is not a startup,” notes Narasimha.

Open-sourcing alongside IoT

Aeroshil Nameirakpam, Managing Director, Nibiaa Devices Pvt Ltd, notes that taking the open source route might just be the key to skyrocket India as an IoT champion. He says, “The concept of open source hardware development is coming along and it is creating a revolution which actually happened during the software era during the early 21st century where open sourcing of codes actually helped establish a lot of new companies.”

In fact, IoT is arguably the right contender to be brought about in this new age where collaboration, a quintessential part of the IoT space,

is done to build scalable and efficient products. Nameirakpam says, “When it comes to building sensors and devices, countries which are very good at it, like China and Taiwan, are helping other companies. When it comes to application building, which we Indians are very good at, we can start thinking of providing vertical solutions.”

Becoming part of the open source community can be a gateway for Indian products or solutions to reach a wider audience globally. A lot of hardware libraries and packages are readily available right now on communities like GitHub who are constantly trying to bridge the gap between emerging technologies through collaboration, which, if we start developing on our own, would take a lot of time. If used judiciously, the open source phenomenon could be the backbone upon which Indian IoT startups thrive.

“Every country is trying to bring about this disruptive technology and we should participate in it and take more of a big leap because we have the resources and a huge pool of educated, smart youngsters who are looking for opportunities,” concludes Nameirakpam.

The future of IoT in India

India has been the beacon of the software revolution that took place in the early 2000s. Software startups got their due in the IT industry when the country welcomed it with open arms. However, the same cannot be said for hardware startups that have seldom been able to mimic similar success stories. “India is strong in software but there is a significant lack of skills

in terms of hardware. India needs to strengthen its manufacturing capability if it wants to take on the big guys like China,” says Bandyopadhyay.

Getting funds for any startup is an uphill task, more so for an electronics startup. But IoT startups get a certain leeway mainly due to the availability of a software component to it, which proves to a potential investor that it has the capability to be scaled to new, greater heights.

Nameirakpam argues that only building the software component of IoT while sidelining the other core components will not lead to a holistic development of the ecosystem. He says, “I think for an overall ecosystem development we need an all-round development. We have to give emphasis to hardware startups as well.”

An example he gives relates to the dire situation the country faced during the havoc caused by the second wave of the Covid-19 pandemic in terms of fulfilling the requirements of essential medical equipment like pulse oximeters. “People needed to save lives on the ground, and we were importing them from other countries. If there were startups that were making these medical devices in the country, and if they got the required funding, there is a possibility that we would have been able to procure them faster.”

“The truth is, when you make a manufacturing capability stronger, the entire value chain benefits from it. It can push the startup culture beyond the top ten cities that we are currently looking at, and it can go into the smaller towns too,” adds Panneerselvam Madanagopal, Advisor at Niti Ayog and Senior Advisor at T-Hub.

In addition to this, telecom—a core element of the IoT segment—is also witnessing a new revolution with the introduction of 5G in the country, which many believe would be a gamechanger for IoT players. In spite of that, the maturity of the entire ecosystem seems to be a far cry. Even with 4G, different parts of the country have different quality of network access, which broods the question: Will 5G really be the game-changer one expects it to be?

“It is not just a matter of a few isolated companies becoming efficient, but the overall IoT ecosystem must